



V.O.I.C.E.

Virtual Observation and Interactive Clinical Education

Emotionally Expressive, Agentic Multimodal Virtual Patients for Clinical Simulation

Yujing Wang¹, Mingkai Gao¹, Brandon Watanabe, Zhuoying (Ellen) Xue¹, Yuqi Hu, Zhenzhen Qin¹,
Fuchao (Rick) Yang¹, Kunyi Shi¹, Shreevidhya Shambanna¹, Professor Ilmi Yoon¹

¹Northeastern University, Silicon Valley



Poster #32

Background

Speech-language pathology (SLP) training prepares students for **clinical communication and decision-making** before clinical placement. However, opportunities for **realistic practice** remain limited. Standardized patients are interactive but **resource-intensive** and difficult to scale, while text- and video-based simulation tools are more accessible and repeatable, but do not support **active, real-time verbal interaction**.

V.O.I.C.E. addresses this gap through an **agentic, browser-based** virtual patient system that enables **natural spoken interaction**. It uses **LLM-driven dialogue** and **multimodal expression** to produce context-aware, emotionally expressive responses, while remaining **scalable and accessible** for repeated practice. This allows learners to actively engage in clinical communication and decision-making, and better prepare for real clinical encounters.

Simulation Interface

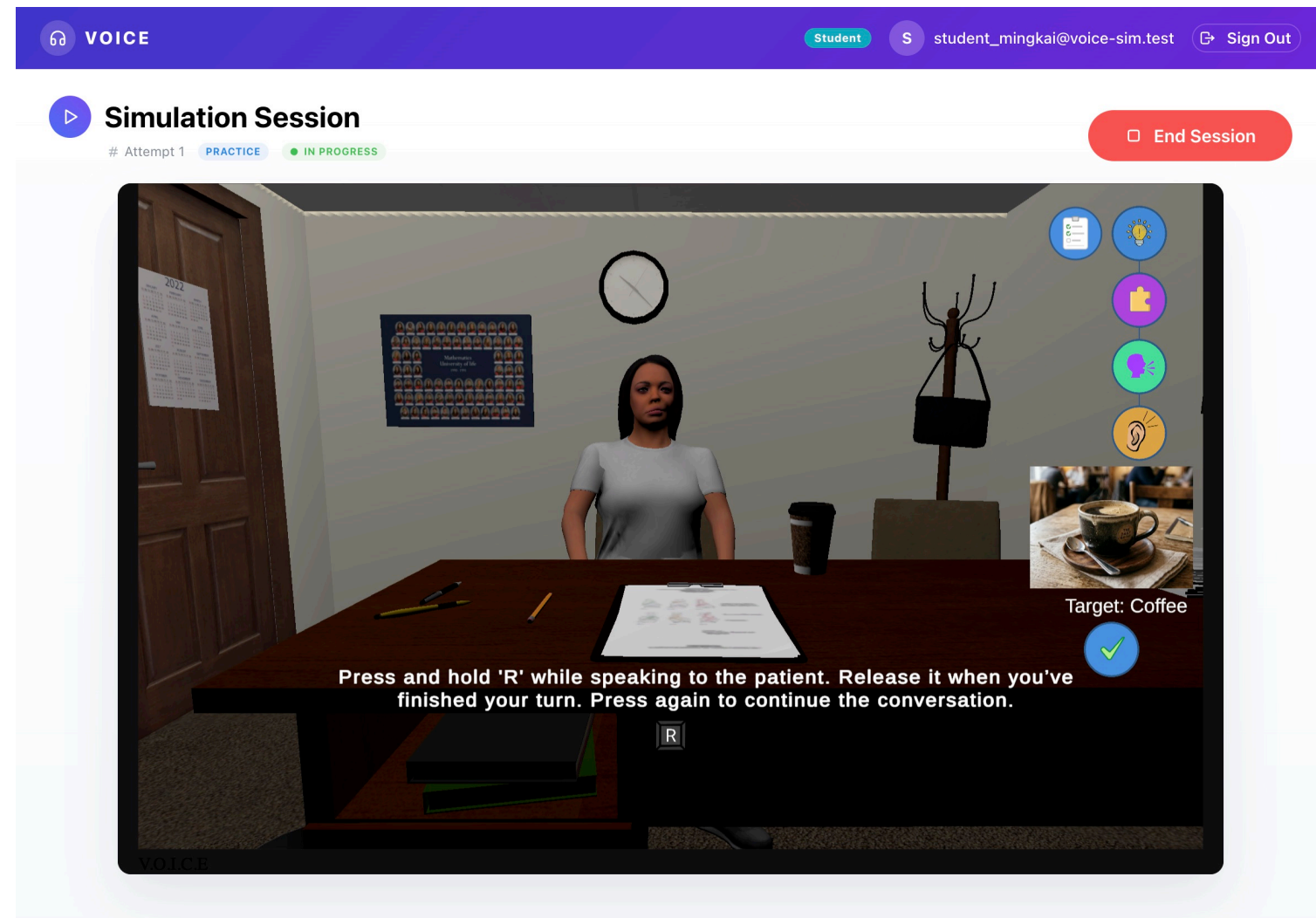


Figure 1. Unity WebGL simulation interface showing push-to-talk (R-key) interaction, emotionally expressive synchronized behavior, and integrated clinical scaffolding through task checklists, cues, and real-time feedback.

System Architecture

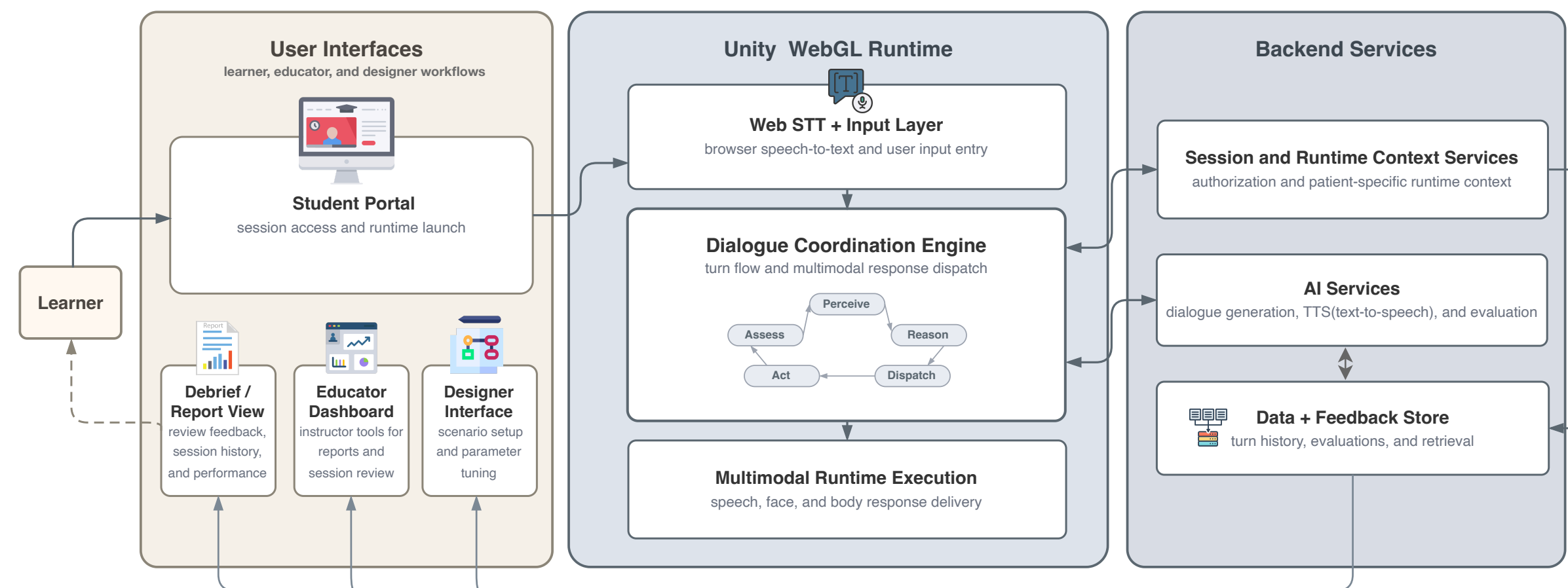


Figure 2. System-level overview of the V.O.I.C.E. platform, illustrating web-based interfaces, Unity WebGL runtime, and cloud-backed services.

Key Highlights

Elevating Clinical Preparedness

Bridges the transition from **classroom theory** to **real-world clinical interaction** through accessible, emotionally realistic practice. V.O.I.C.E. helps learners build **confidence, communication fluency, and readiness** through repeated, safe-to-fail clinical exposure, with early user study results supporting this pedagogical approach.

Multimodal Patient Simulation

Expressive, agentic responses synchronize **speech, facial expression, body motion, and lip sync** to create immersive clinical realism. Through a **Unity WebGL interface** with push-to-talk interaction, the system delivers naturalistic audio-visual feedback that better reflects real patient encounters.

Scalable Classroom-Ready Deployment

A distributed, **browser-based** system designed for direct integration into SLP coursework. Developed with faculty input, it supports instructional **clinical scaffolding** through task checklists, real-time feedback, **AI-supported debriefing**, and performance analytics to support consistent, high-volume learning-by-doing environments.

Multi-Agent Ecosystem

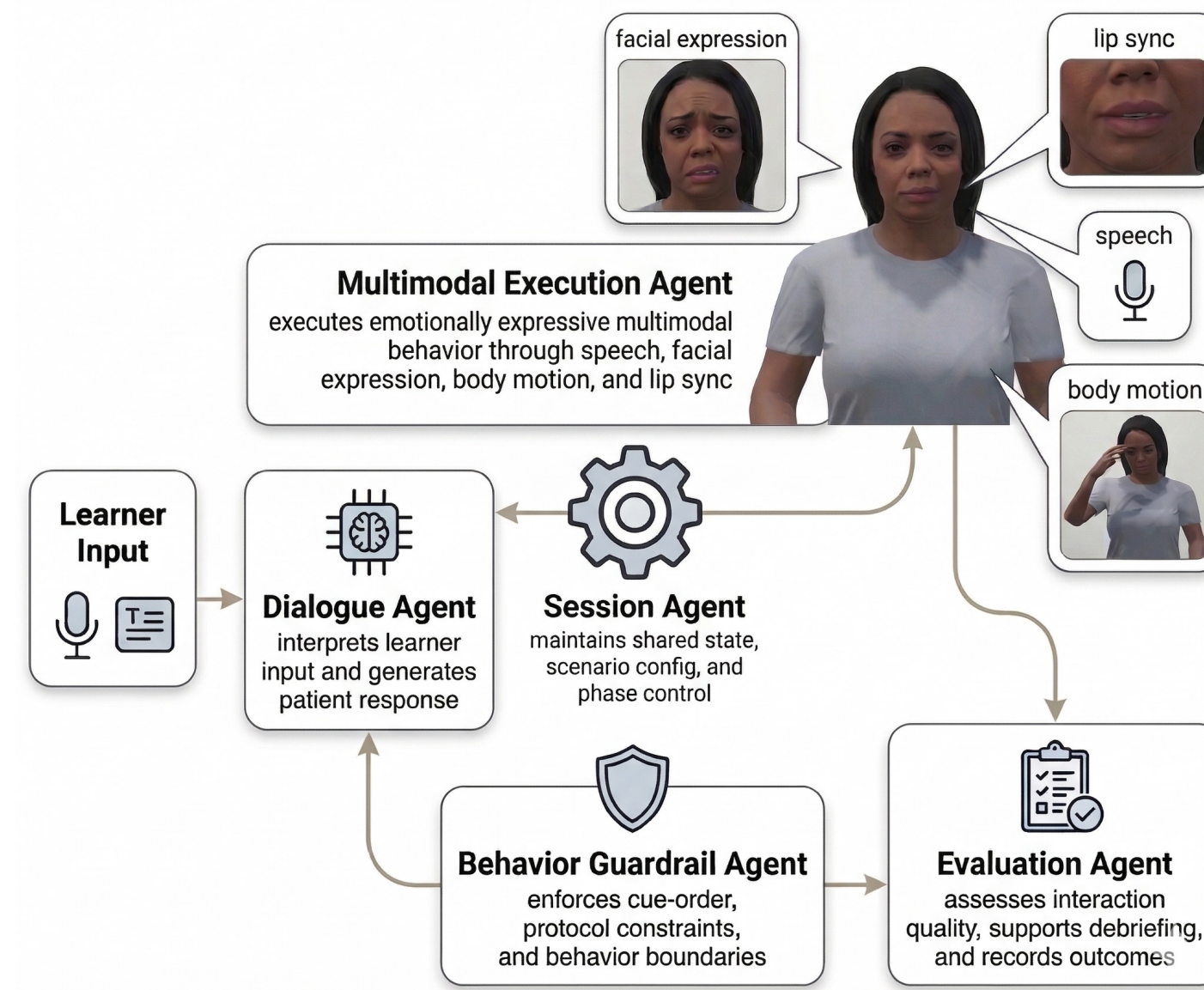


Figure 3. Agent-level view of how coordinated agents transform learner input into emotionally expressive multimodal patient behavior.

Multimodal Animation Pipeline

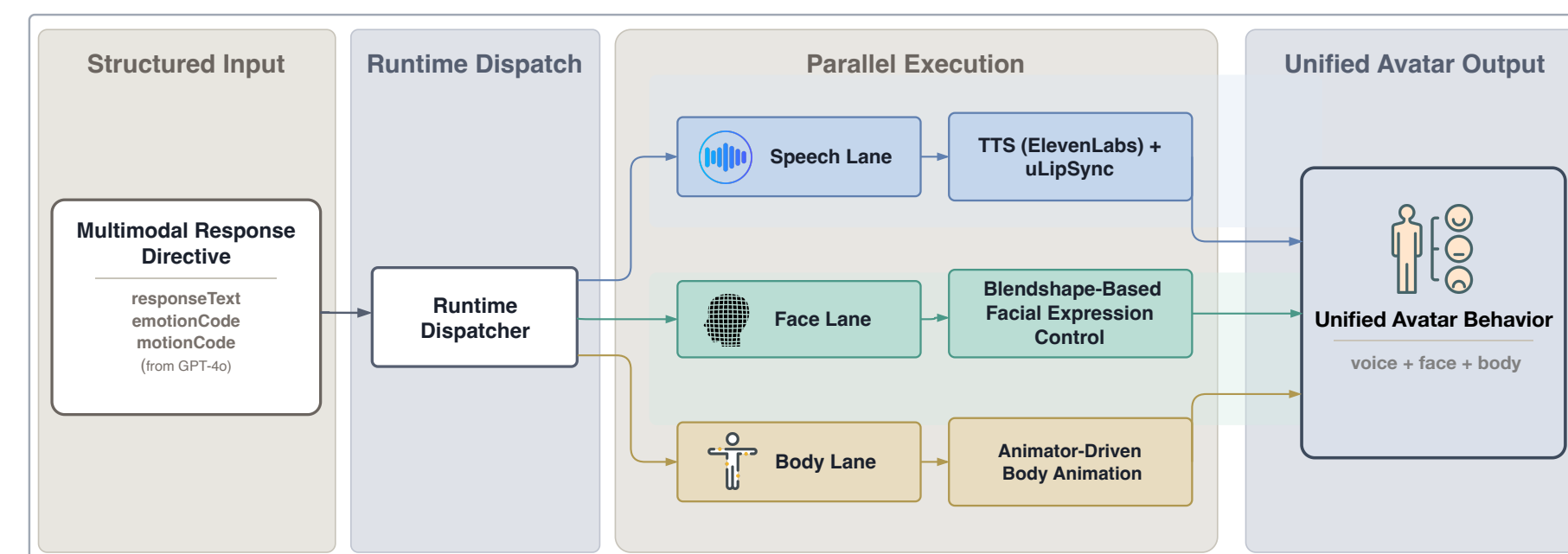


Figure 4. Execution-level view of how structured response signals are dispatched across speech, facial, and body channels and realized as unified avatar behavior.

Evaluation & Educational Impact

In a within-subjects study with **11 graduate SLP students**, confidence increased significantly after using V.O.I.C.E. (composite **+1.09** on a 6-point scale, $p = .032$), and higher **perceived realism** was associated with larger gains.

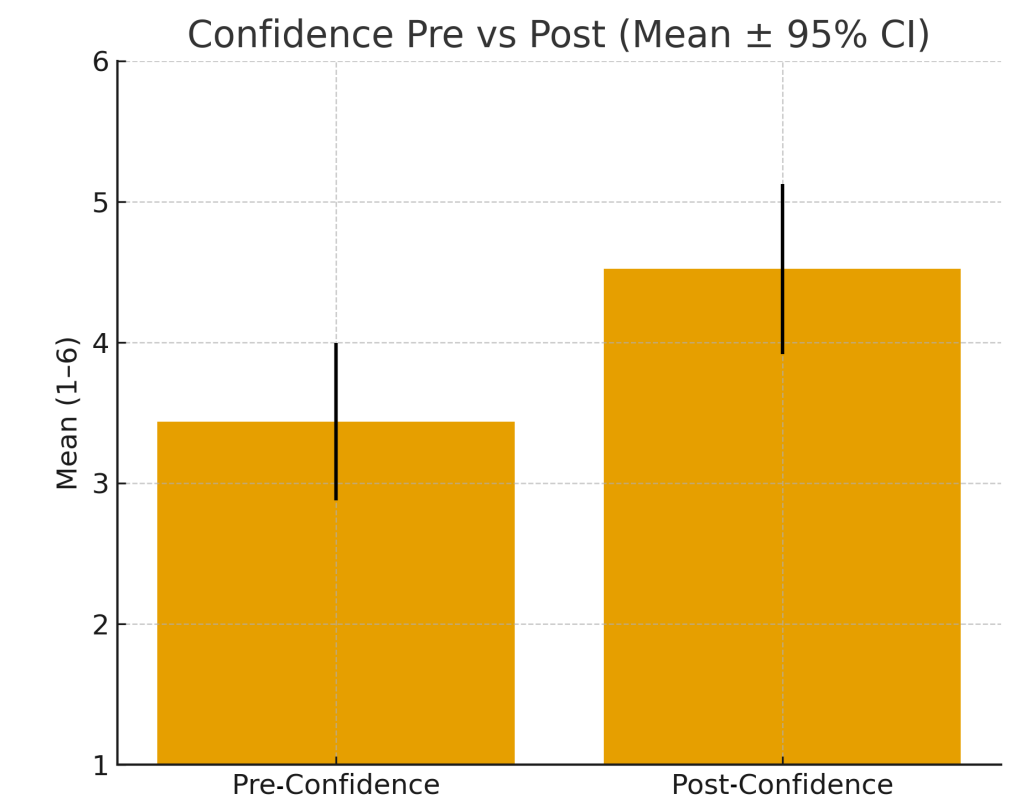


Figure 5. Pre-post confidence ($p = .032$).

Beyond measured confidence gains, early faculty feedback and planned classroom adoption suggest strong educational relevance.

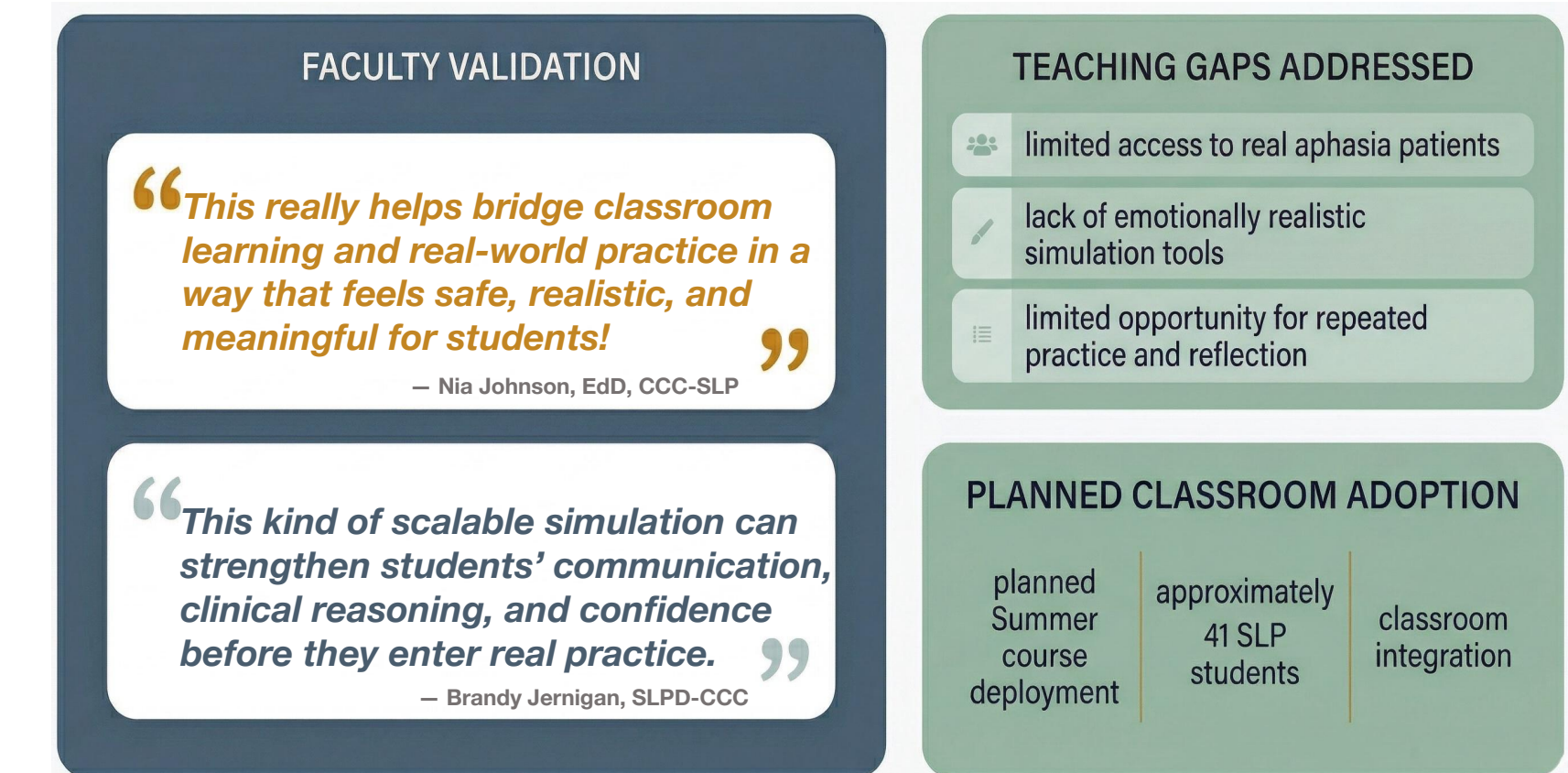


Figure 6. Faculty perspectives, teaching gaps addressed, and planned classroom integration, including summer deployment in SLPA 6415 (Speech-Language Pathology Advanced Clinical Practicum 1).

Conclusion & Future Directions

V.O.I.C.E. shows that **emotionally expressive, browser-based** virtual patients can make clinical training more **realistic, repeatable, and scalable**. Rather than a scripted, static simulation, it demonstrates an **agentic multimodal system** designed for **real educational use**. More broadly, it points toward new models for **ethical, technology-enhanced clinical training** that can better prepare future clinicians for **patient-centered care** in increasingly complex healthcare environments.

Future directions: Adapting the session-agent framework for **broader teaching scenarios**, and exploring public-facing **translational use** for communication support among **stroke survivors, caregivers, and families**.